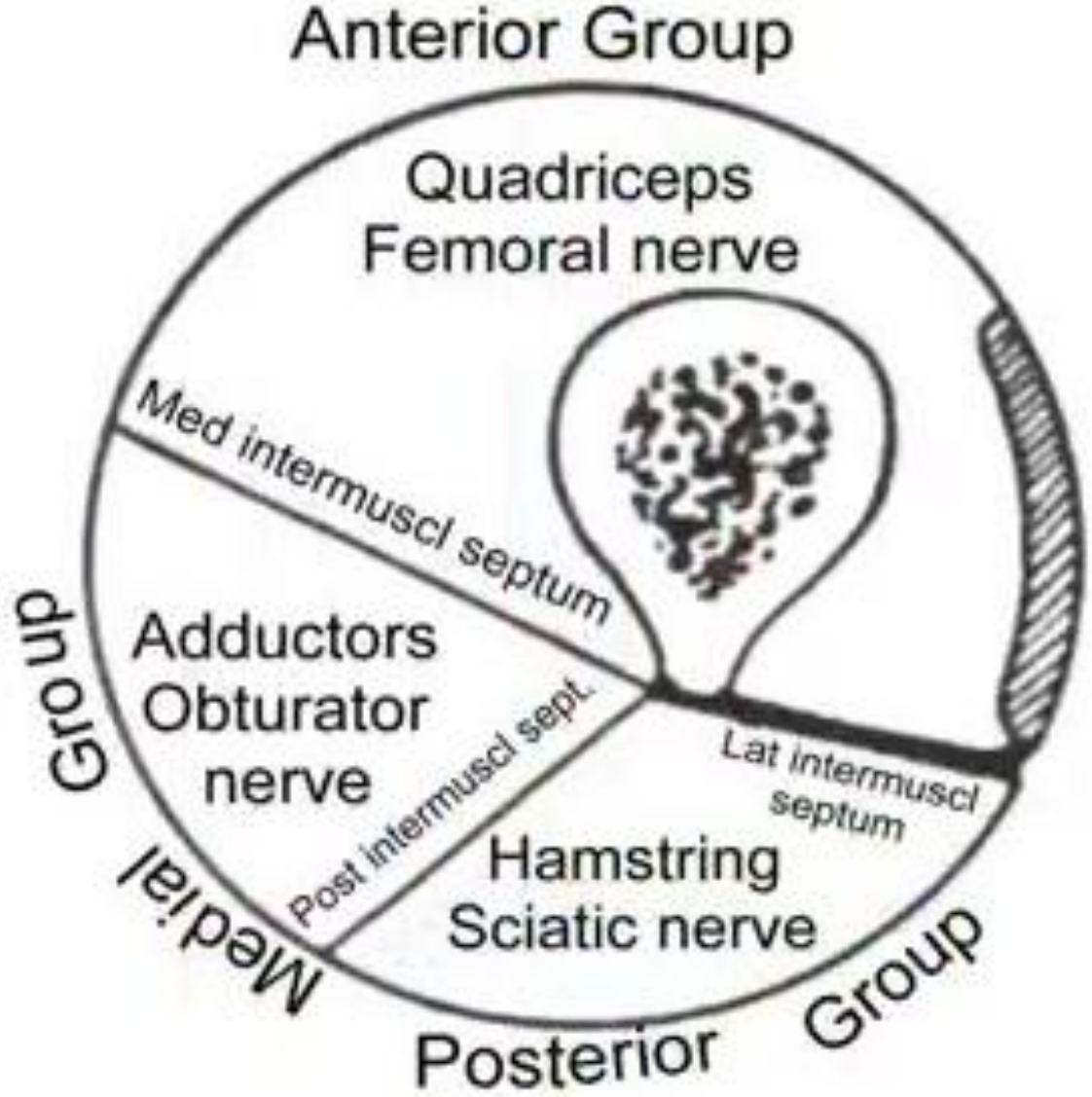
Compartments of the THIGH

DR OYESHIKU

- Anterior

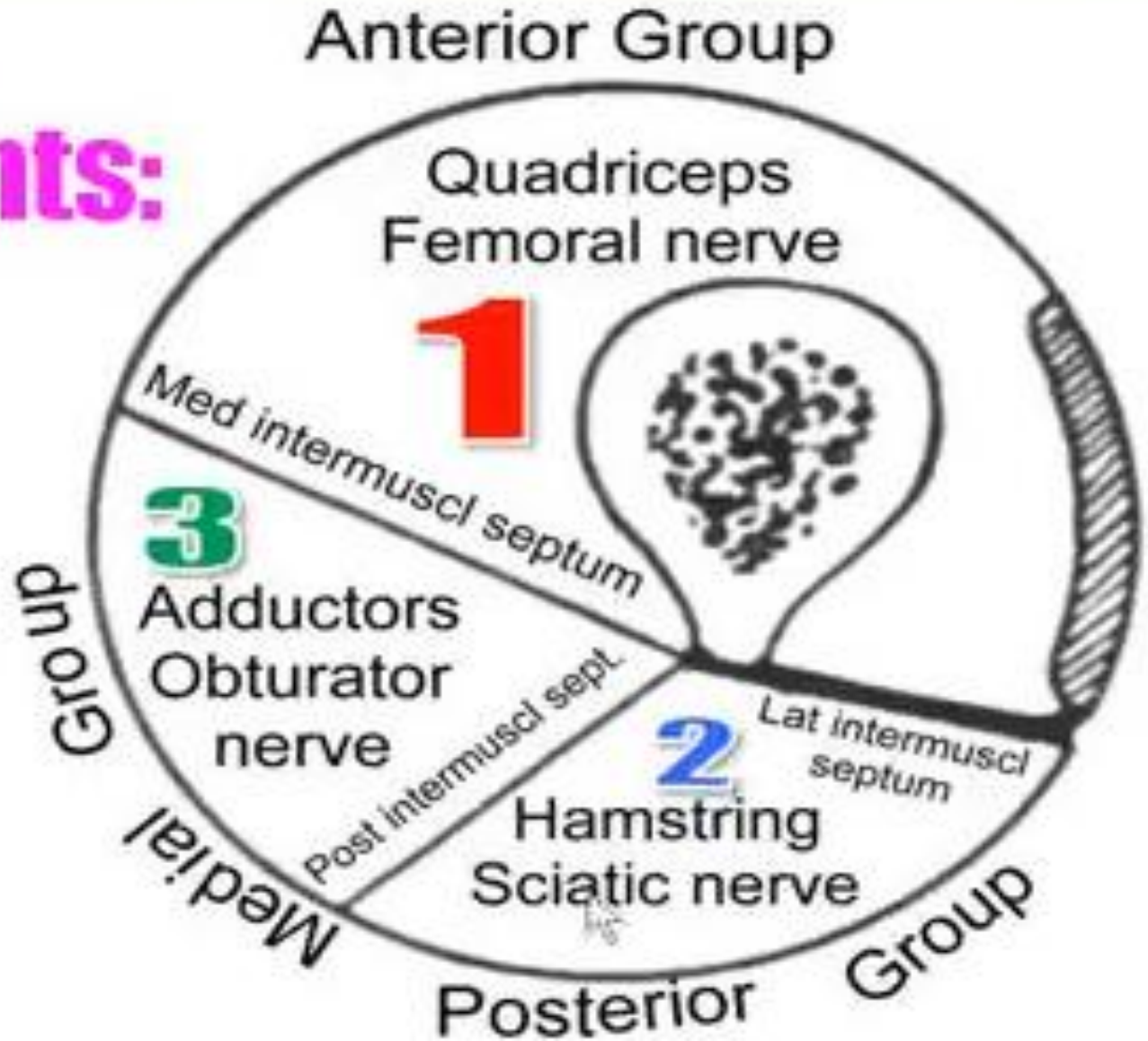
ORGANIZATION OF THE THIGH



ORGANIZATION OF THE THIGH

3 Compartments:

1. Anterior
2. Posterior
3. Medial



Compartment	Muscle Group	Nerve Supply	Arterial Supply	Main Action
Anterior	Quadriceps femoris	Femoral nerve	Femoral artery	**Extension** of the knee joint
Posterior	Hamstring muscles	Sciatic nerve	4 perforating arteries	1. **Extension** of hip joint 2. **Flexion** of knee joint
Medial	Adductors	Obturator nerve	Obturator & profunda femoris arteries	**Adduction** of the hip joint

Nerve Supply of the Skin and Dermatomes: 1 – 2 - 3

- **Upper Part: L1**

  femoral branch of the genitofemoral nerve

- **Middle Part: L2**

  lateral & intermediate cutaneous nerves of the thigh

- **Lower Part: L3**

  lateral & intermediate cutaneous nerves of the thigh



Fascia of the Thigh

Superficial Fascia:

1. **Fatty layer:** contains **GSV**, **LN** & **cutaneous nerves**
2. **Membranous layer**

Deep Fascia: fascia lata:

@ **Forms** iliotibial tract (lat side)

@ **Contains** saphenous opening



Action of the Muscles

Quadriceps: 1, 2, 3 & 4

- Extension of knee … RF: flexion of hip

Sartorius

- Flexion of hip + Flexion of knee joint

Pectineus

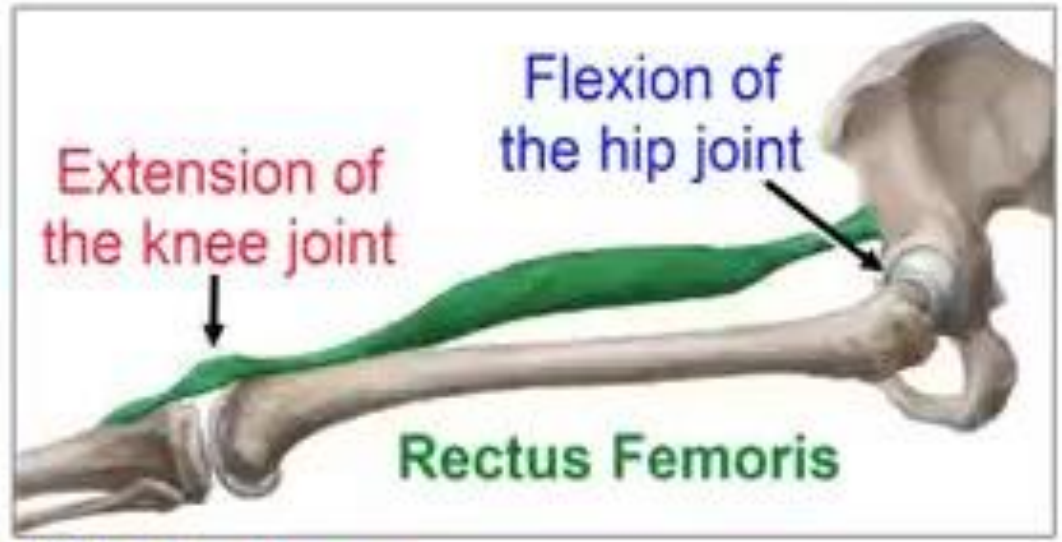
- Flexion & adduction of hip joint

Iliopsoas

- Main flexor of the hip joint



Front of the Thigh





Psoas major is the main flexor of the hip joint

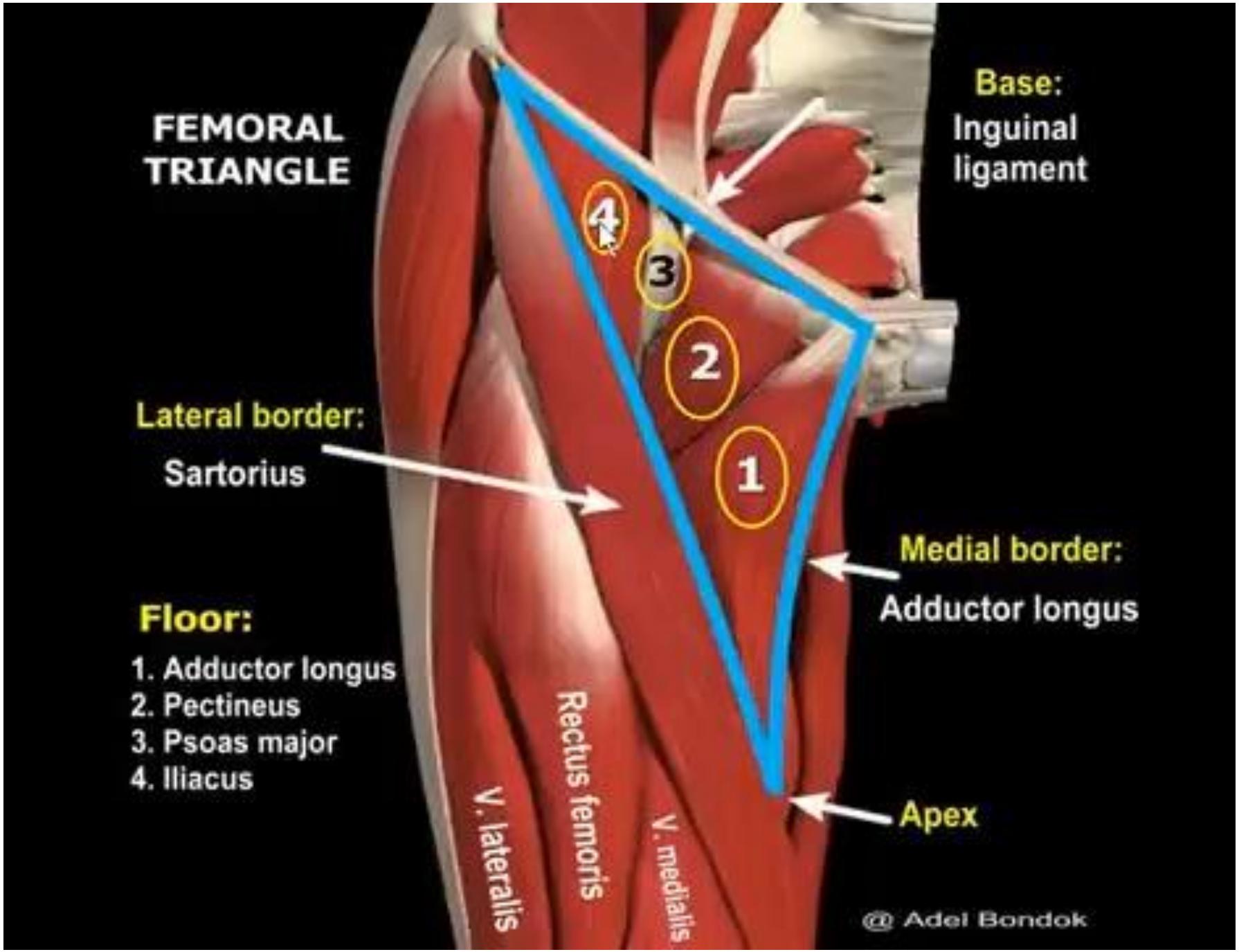


FEMORAL TRIANGLE

1. Position
2. Boundaries
   - Medial wall
   - Lateral wall
   - Base
   - Apex
   - Roof
   - Floor
3. Contents






FEMORAL TRIANGLE
Base:
Inguinal ligament
4
3
2
1
Lateral border:
Sartorius
Medial border:
Adductor longus
Floor:
1. Adductor longus
2. Pectineus
3. Psoas major
4. Iliacus
Apex
V. lateralis
Rectus femoris
V. medialis
@ Adel Bondok

CONTENTS OF FEMORAL TRIANGLE

1. Femoral Sheath
2. Femoral Vessels:
   a. Femoral artery
   b. Femoral vein
3. Nerves:
   a. Femoral Nerve
   b. Femoral branch of genitofemoral nerve
   c. Sympathetic plexus around FA
4. Deep Inguinal Lymph Nodes



FEMORAL SHEATH

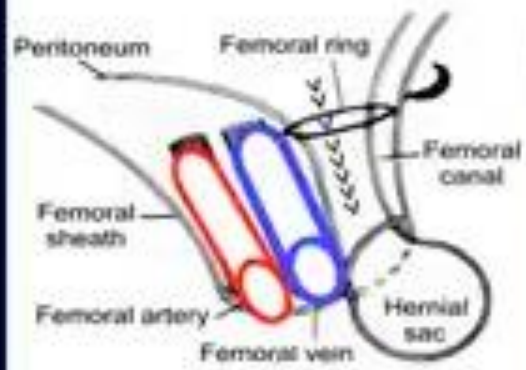
Walls:

Anterior wall:

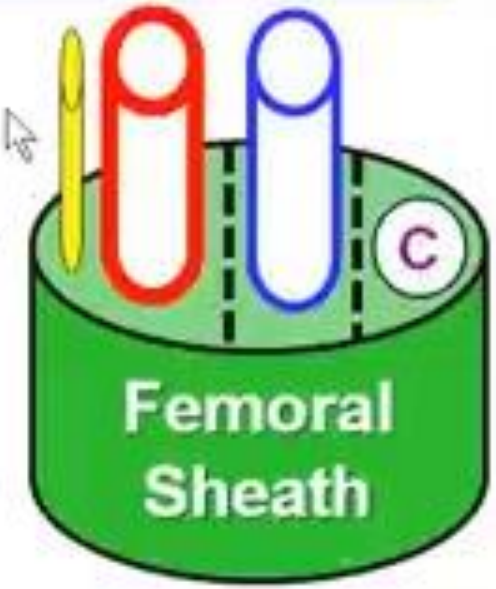
Posterior wall:

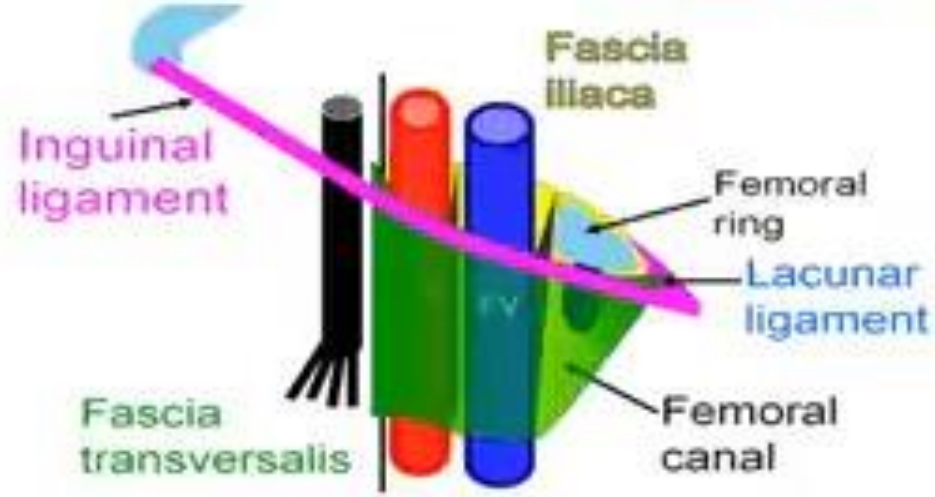
Compartments: 3

1. **Lateral:** FA + FGFN
2. **Middle:** FV
3. **Medial:** femoral canal

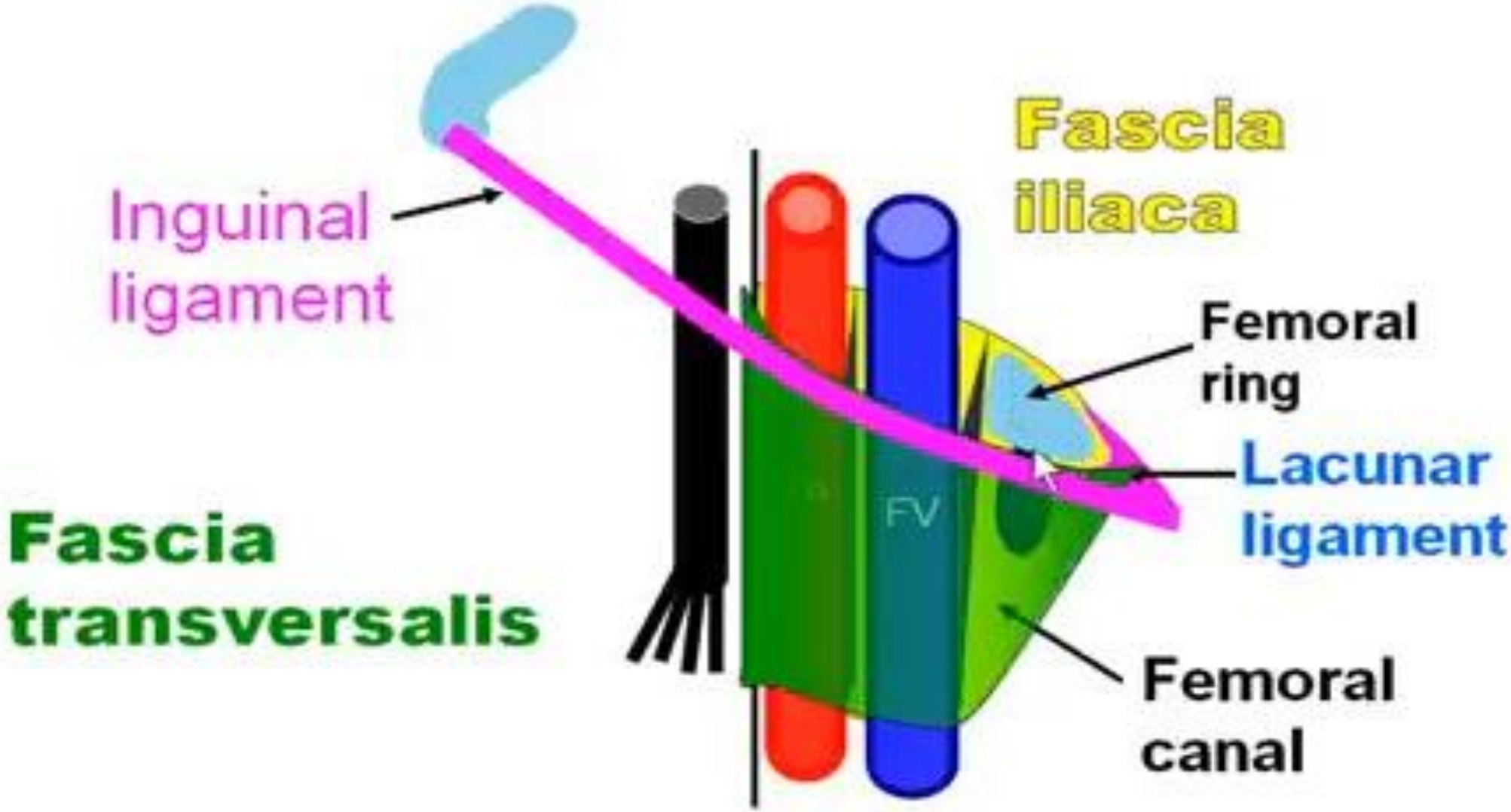


Medial compartment (femoral canal) of the femoral sheath and the passage of the femoral hernial sac





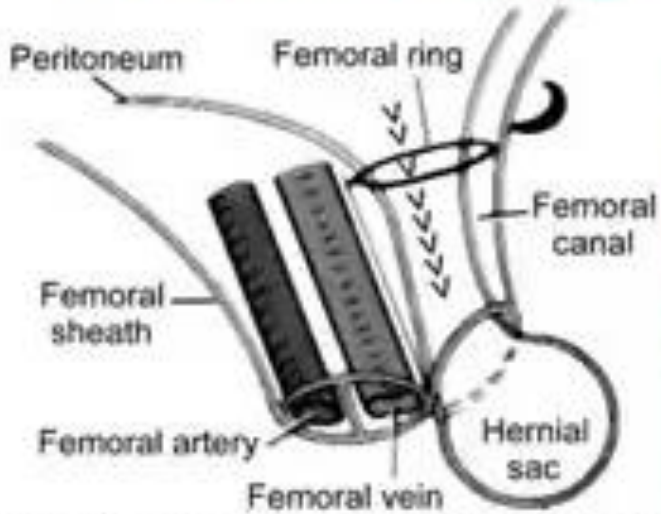
FEMORAL SHEATH



FEMORAL CANAL

- **Femoral Ring:** bounded by
- **Contents**
- **Advantage**
- **Disadvantage**





Medial compartment (femoral canal) of the femoral sheath and the passage of the femoral hernial sac



- Medial

Medial Compartment

Muscles:

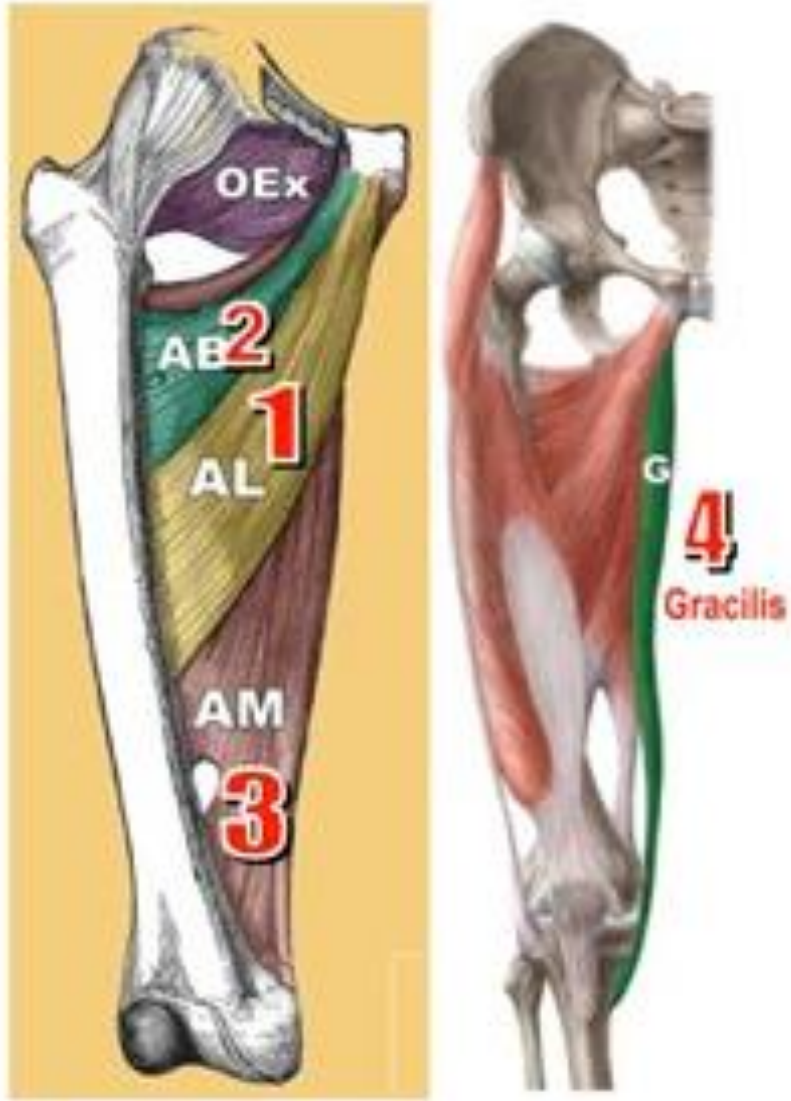
1. Adductor longus
2. Adductor brevis
3. Adductor magnus
4. Gracilis

Artery: obturator artery

Nerve: obturator nerve

Action of Addctors:

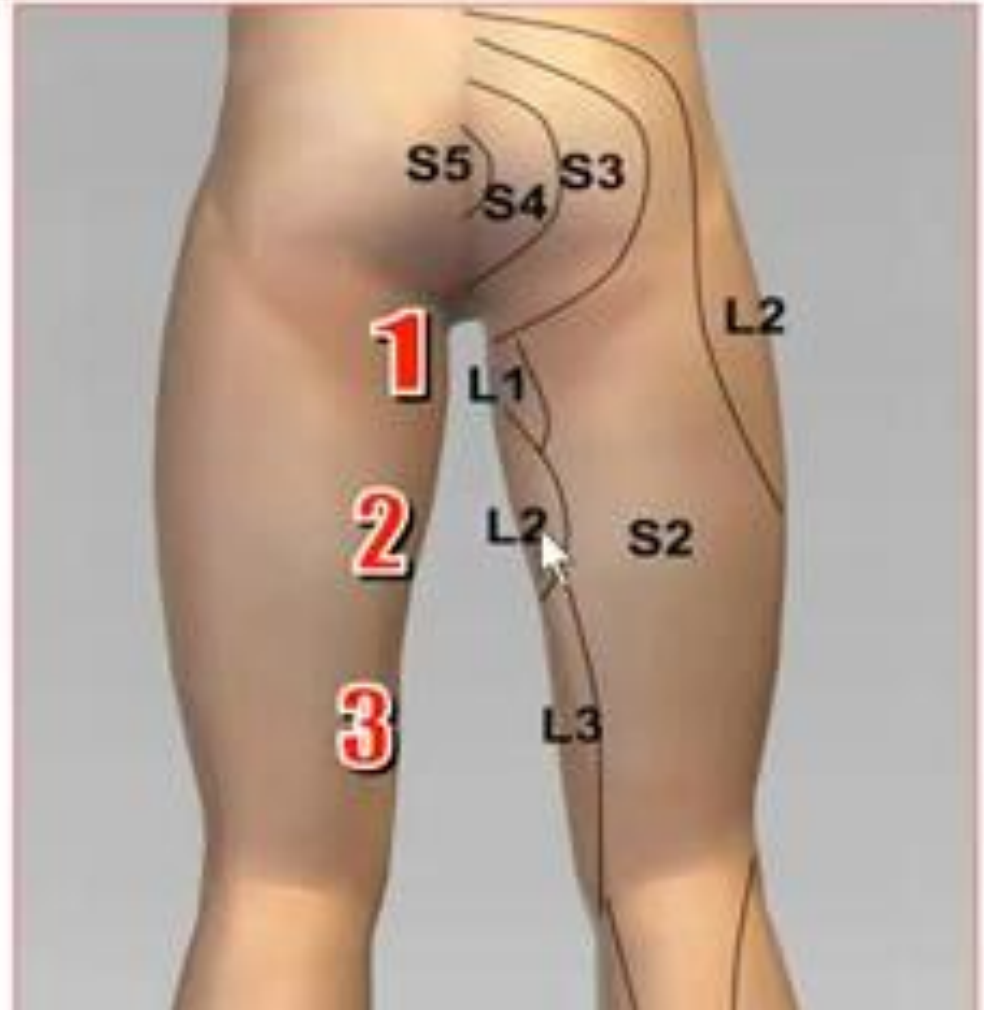
Adduction, flexion & lateral rotation



Nerve Supply of the Skin and Dermatomes: 1 – 2 - 3

MEDIAL SIDE

- Upper Part: L1
- Middle Part: L2
- Lower Part: L3



Adductor Muscles

Origin:

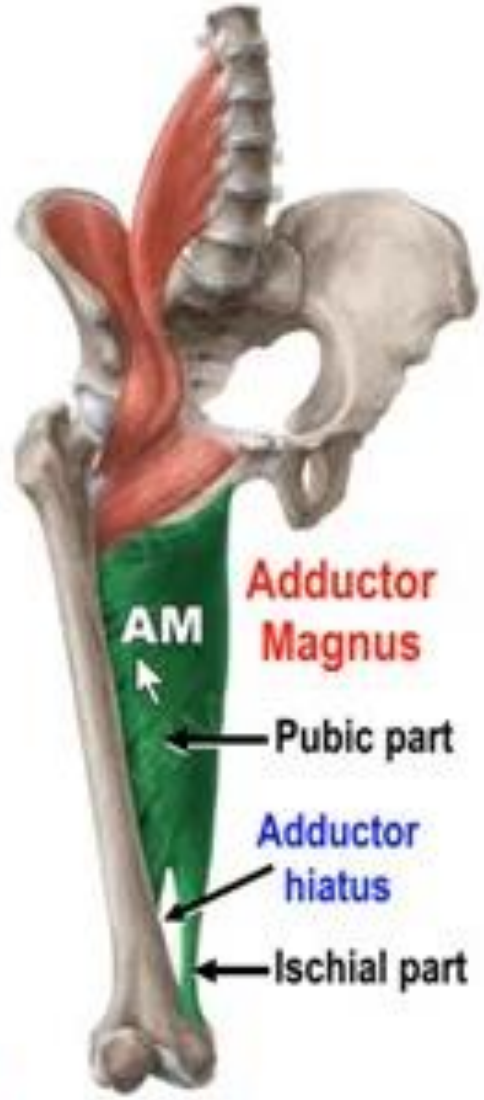
Insertion:

Nerve Supply:

Action:









Action of the Adductor Muscles

Adduction

Flexion

Lateral Rotation



Lateral Rotation





Cross Your Legs

Adductor Canal
"Subsartorial Canal"

Location:

Begins at:

Ends at:

Boundaries





ADDUCTOR CANAL
Subsartorial Canal

Location

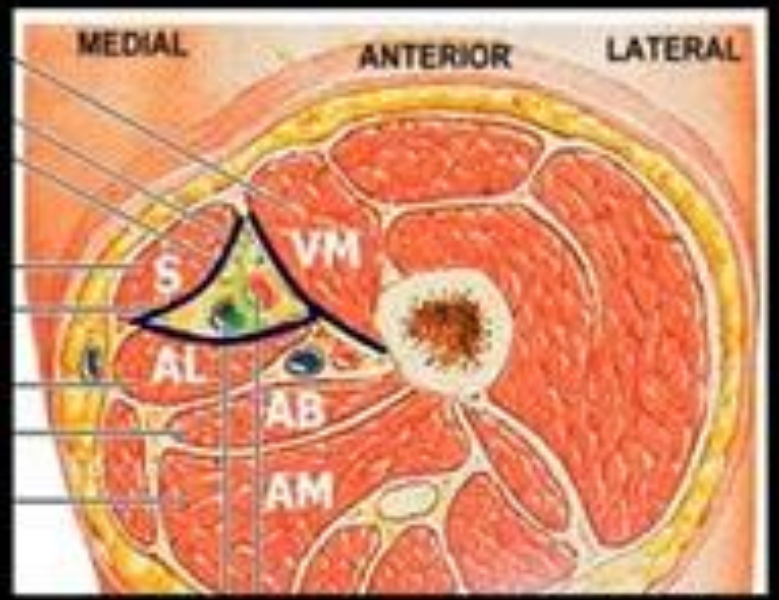
Boundaries

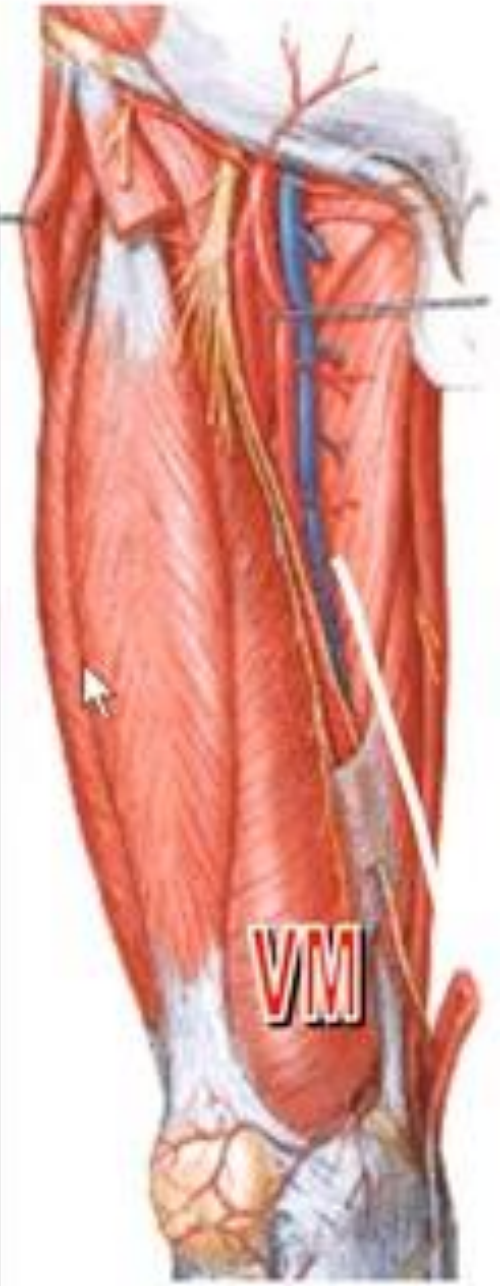
Anteromedial S

Anterolateral VM

Posterior: AL AM

Contents:





ADDUCTOR CANAL
Subsartorial Canal

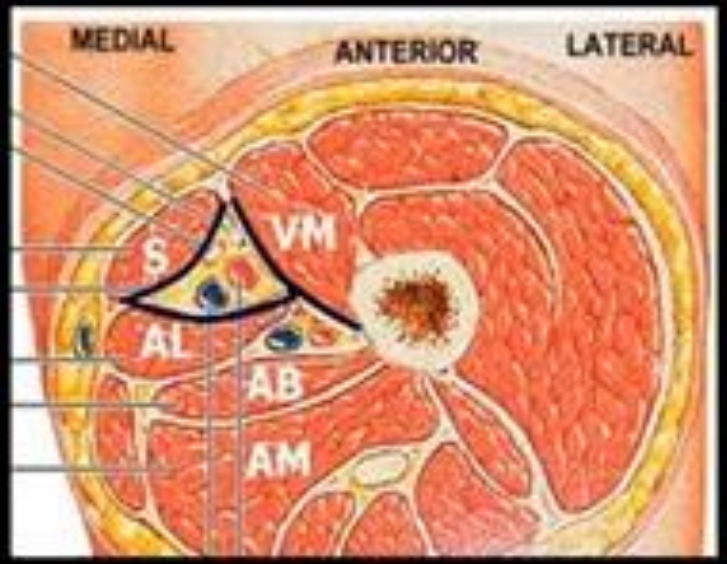
Location

Boundaries

Anteromedial S

Anterolateral VM

Posterior: AL AM



Contents:

Vessels:

1. Femoral Artery & descending genicular artery
2. Femoral Vein

Nerves:

1. Saphenous
2. Nerve to vastus medialis
3. Subsartorial plexus



posterior

anterolateral



anteromedial

Adductor longus &
Adductor magnus

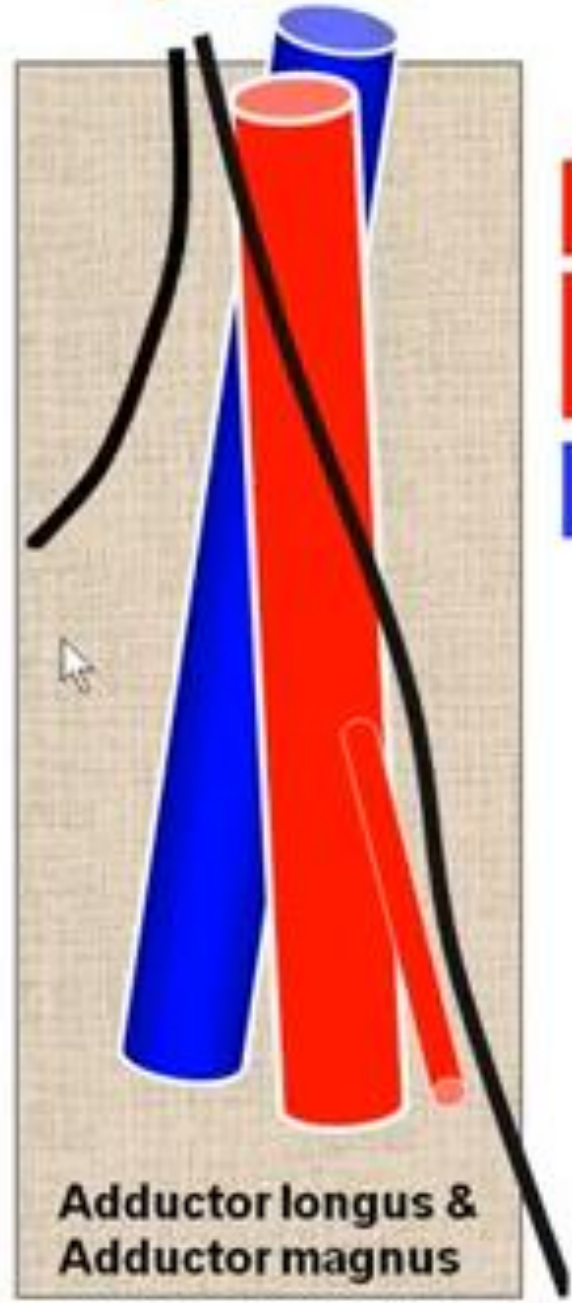
ADDUCTOR CANAL
Subsartorial Canal

posterior

Saphenous nerve

Nerve to Vastus medialis

anterolateral



Femoral artery

Descending genicular artery

Femoral vein

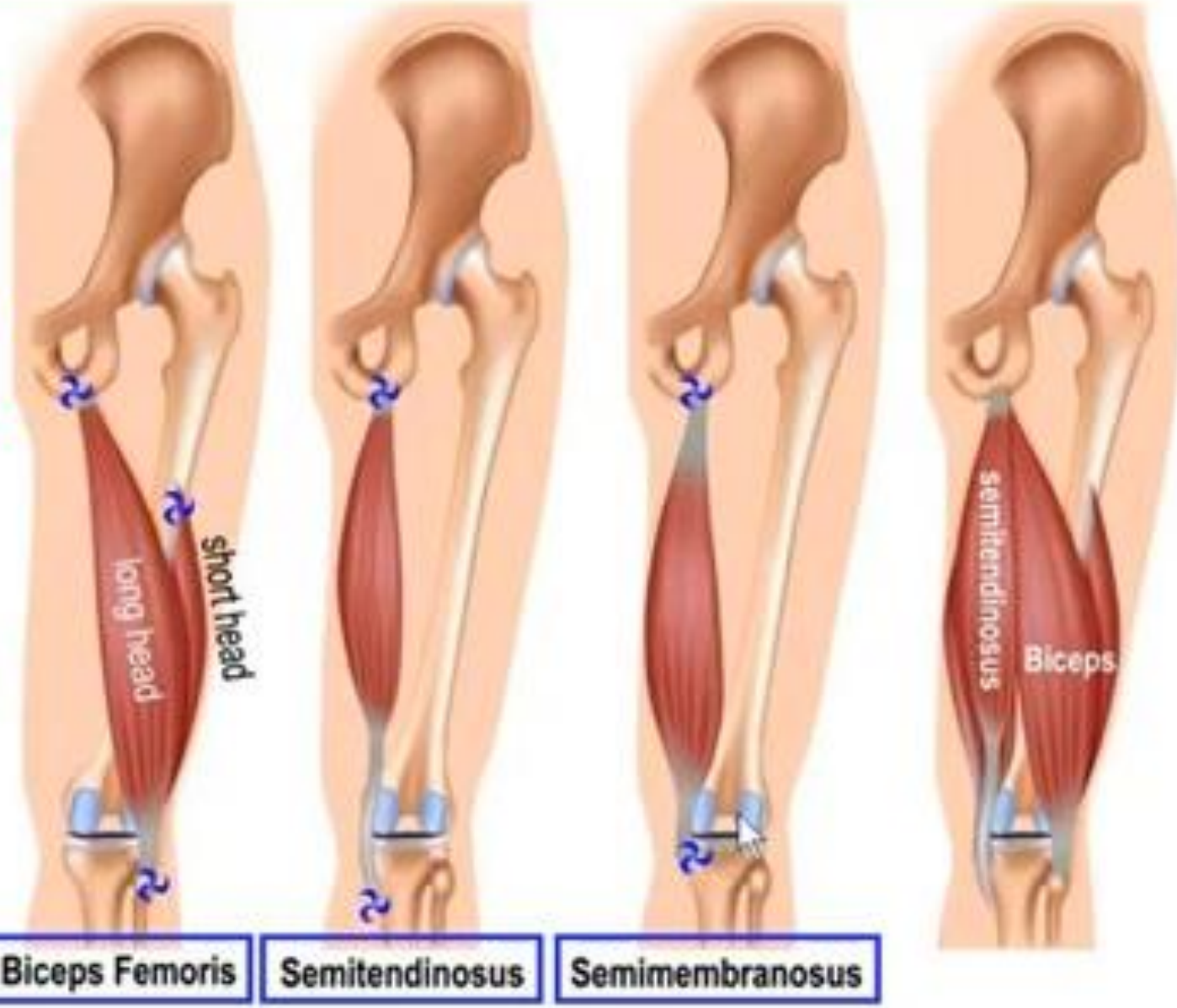
anteromedial

ADDUCTOR CANAL
Subsartorial Canal

- Posterior

HAMSTRING MUSCLES

- Common Origin
- Insertion
- Common Nerve Supply
- Common Action



Sciatic Nerve

Origin: L4, 5, S1, 2, 3

Landmarks / Course:

- **Passes in GSF** below the piriformis
- **Covered by** gluteus max & hamstring
- **May be injured by** wrong IM injection

Termination: in middle of back of the thigh

1. Tibial nerve
2. Common peroneal nerve

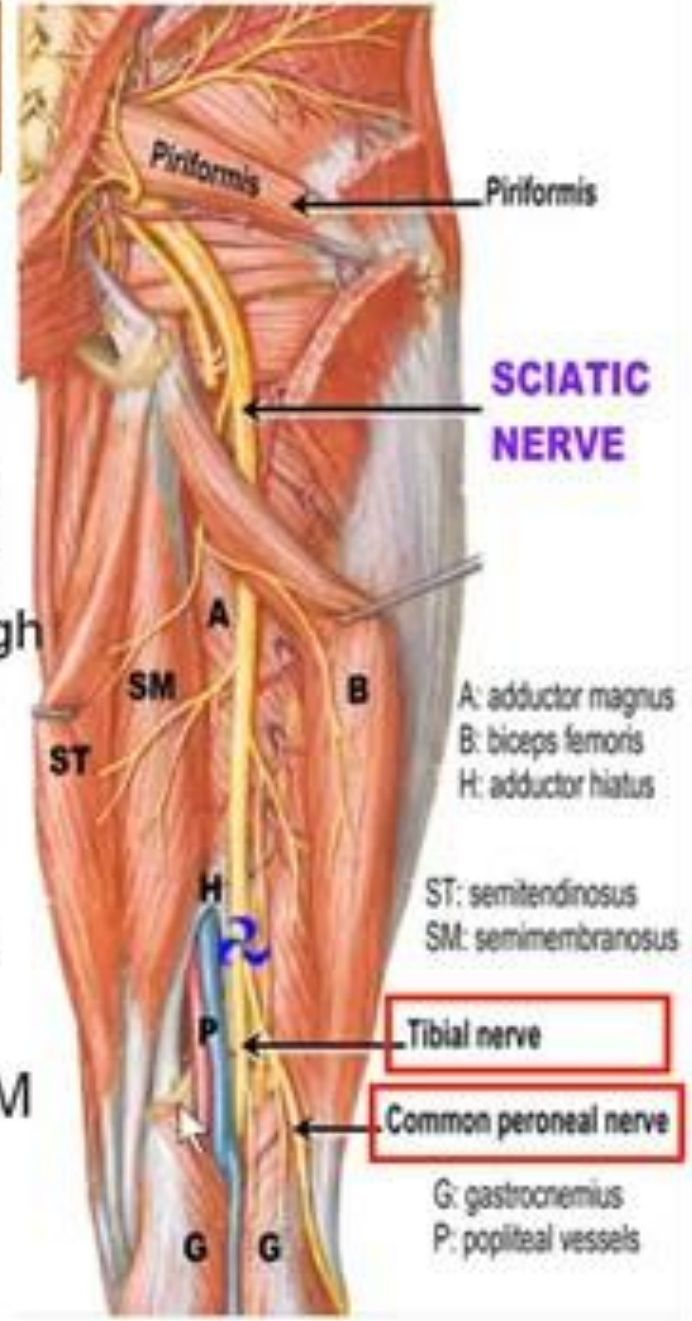
Branches:

1. **Muscular branches:** hamstring muscles

   Biceps, Semitendinosus, Semimembranosus & Ischial part of AM

2. **Articular branches:** to hip & knee joint

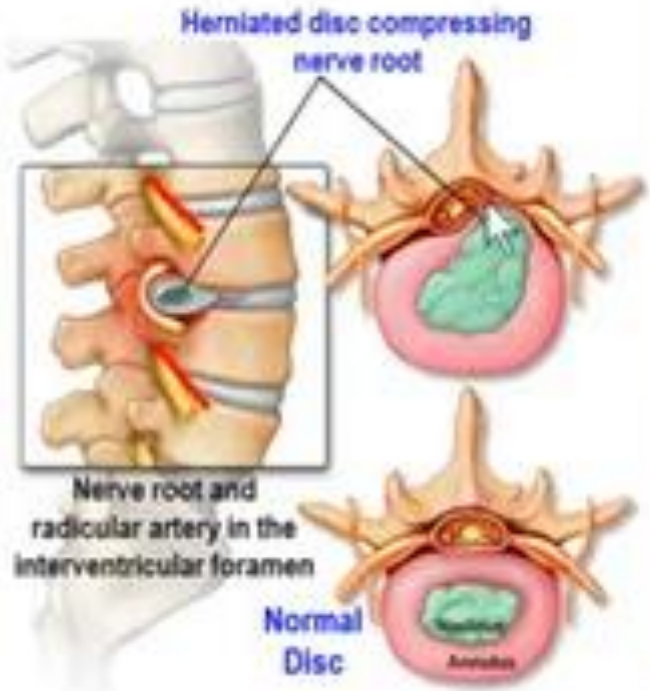
3. 2 Terminal branches



Sciatica

Pain along the sensory distribution of the sciatic nerve

Due to compression of nerve roots by herniated intervertebral disc



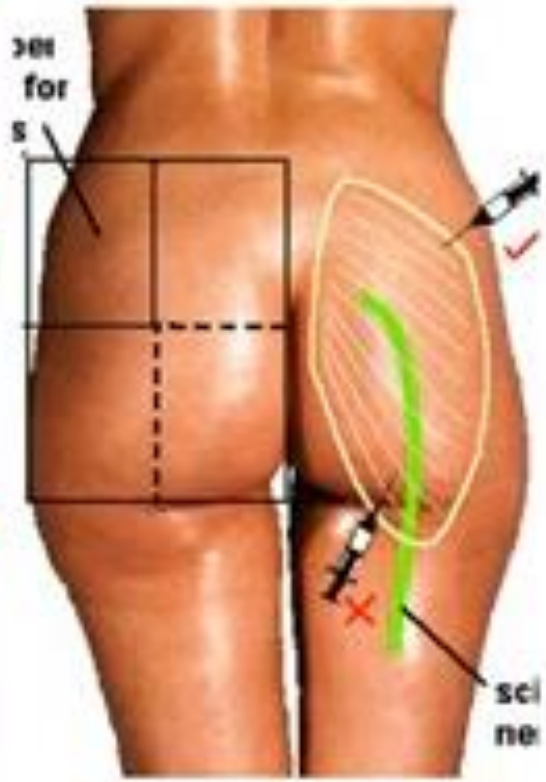




Sciatic Nerve Injury

@ Due to wrong intramuscular injection

@ May be compressed by the piriformis



It causes:

1. **Parakysis of** the hamstring muscles and muscles of the leg & foot.
2. **Loss of sensation** below the knee except the medial side of the leg & foot



Deformity: Foot drop